INCH-POUND
MS39202G
30 August 2012
SUPERSEDING
MS39202F
1 February 2002

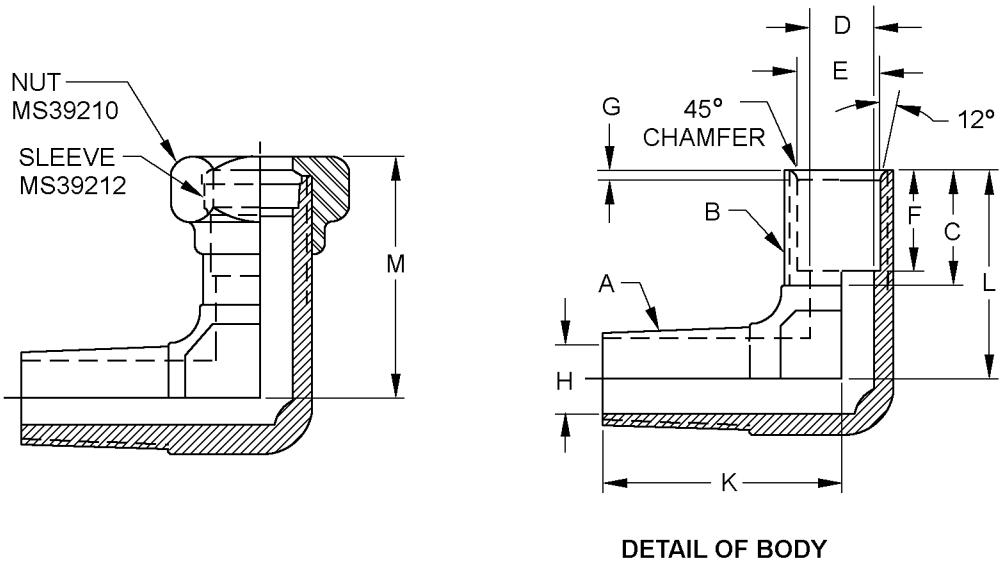
DETAIL SPECIFICATION SHEET

ELBOWS, PIPE TO TUBE – SAFETY SLEEVE COMPRESSION TYPE, 90 DEGREE, MALE PIPE END

Inactive for new design after 29 December 1967.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



DETAIL OF BODY

FIGURE 1. Elbows, pipe to tube – safety sleeve compression type, 90 degree, male pipe end.

AMSC N/A

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:48Z
Check the source to verify that this is the current version before use.

TABLE I. Elbows, pipe to tube-safety sleeve compression type, 90 degree, male pipe end. 1/ 2/

Dash No.	Tube O.D.	A Pipe Thread NPTF	B	C Thread Length	D	E	F	G	H
1	1/8	1/8	5/16 24UNF-2A	9/32	3/32	.135 .139	13/64	1/8	3/16
2	3/16	1/8	3/8 24UNF-2A	9/32	1/8	.196 .200	1/4	5/32	3/16
3	1/4	1/8	1/2 20UNF-2A	13/32	13/64	.261 .265	19/64	13/64	3/16
4	5/16	1/8	9/16 18UNF-2A	13/32	13/64	.324 .328	19/64	13/64	3/16
5	1/4	1/4	1/2 20UNF-2A	13/32	13/64	.261 .265	19/64	13/64	9/32
6	3/8	1/4	5/8 18UNF-2A	7/16	9/32	.386 .390	19/64	13/64	9/32
7	3/8	3/8	5/8 18UNF-2A	7/16	9/32	.386 .390	19/64	13/64	27/64
8	1/2	3/8	3/4 16UNF-2A	7/16	27/64	.514 .518	5/16	13/64	27/64
9	5/8	3/8	15/16 16UN-2A	15/32	1/2	.641 .645	11/32	15/64	27/64
10	3/8	1/2	5/8 18UNF-2A	7/16	9/32	.386 .390	19/64	13/64	7/16
11	1/2	1/2	3/4 16UNF-2A	7/16	27/64	.541 .518	5/16	13/64	7/16
12	5/8	1/2	15/16 16UN-2A	15/32	1/2	.641 .645	11/32	15/64	1/2
13	3/4	1/2	1-1/16 16UN-2A	15/32	21/32	.766 .770	11/32	13/64	1/2
14	3/4	3/4	1-1/16 16UN-2A	15/32	21/32	.766 .770	11/32	13/64	23/32
15	7/8	3/4	1-3/16 16UN-2A	15/32	23/32	.891 .895	11/32	13/64	21/32
16	1	3/4	1-5/16 16.UN-2A	15/32	7/8	1.016 1.020	13/32	13/64	23/32
17	7/8	1	1-3/16 16UN-2A	15/32	23/32	.891 .895	11/32	13/64	7/8
18	1	1	1-5/16 16UN-2A	15/32	7/8	1.016 1.020	13/32	13/64	7/8
19	1-1/4	1	1-5/8 16UN-2A	35/64	1-3/32	1.270 1.274	13/32	13/64	7/8
20	1-1/4	1-1/4	1-5/8 16UN-2A	17/32	1-3/32	1.270 1.274	13/32	13/64	1-1/8
21	1-1/2	1-1/2	1-7/8 16UN-2A	9/16	1-11/32	1.520 1.524	15/32	13/64	1-11/32

See notes at end of tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:48Z
Check the source to verify that this is the current version before use.

TABLE I. Elbows, pipe to tube-safety sleeve compression type, 90 degree, male pipe end – Continued. 1/ 2/

K	L	M
3/4	21/32	31/32
3/4	21/32	1
25/32	3/4	1-5/32
25/32	13/16	1-3/16
1-1/16	7/8	1-9/32
1-1/16	15/16	1-3/8
1-3/16	1-1/8	1-9/16
1-3/16	1-1/8	1-21/32
1-3/16	1-5/16	1-27/32
1-7/16	1-1/8	1-9/16
1-1/4	1-3/16	1-23/32
1-7/16	1-5/16	1-27/32
1-9/16	1-1/2	2-3/32
1-9/16	1-1/2	2-3/32
1-3/4	1-3/4	2-11/32
1-3/4	1-3/4	2-11/32
1-15/16	1-3/4	2-11/32
1-15/16	1-3/4	2-11/32
1-5/16	2-1/4	2-7/8
2-5/16	2-1/4	2-7/8
2-9/16	2-3/8	3-1/16

1/ Dimensions are in inches.
2/ This illustration is for identification and is not intended to restrict designs or shapes not dimensioned.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Tolerances: Unless otherwise specified, decimals ±.005, fractions ±1/64, degrees ±2°.

Threads: The threads shall be in accordance with FED-STD-H28.

Elbows. See table I.

Material:

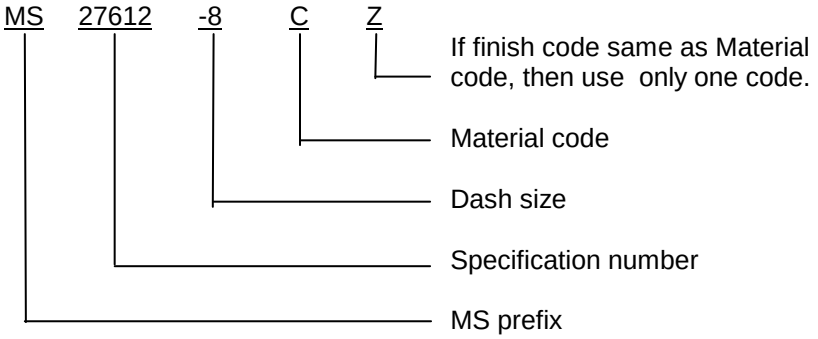
Code	Material
C	Steel bar, type AISI-C-1112, C-1113, C-1117, C-1118 or C-1137 in accordance with SAE AIR4127.
F	Alloy steel, in accordance with SAE AIR4127.
S	Corrosion resistant steel in accordance with SAE AMS5659, SAE AMS5862, or, alloy 15-5 PH in accordance with ASTM A564/A564M type XM-12 or UNS S15500, or SAE AMS5665.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:48Z
Check the source to verify that this is the current version before use.

Chemical finish:

Code	Finish
C	Carbon steel: Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick.
F	Alloy: Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick.
S	Corrosion resistant steel: Passivate in accordance with SAE AMS2700, type 6 or 7.
Z	Steel, alternative chem finish: ASTM B633 type VI, FeZn 25.

Part or Identifying Number (PIN) example: (PIN covers the complete item with nuts and sleeves.)



Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides an alternate material, corrosion resistant steel, and Zinc finish via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. This document reference the following:

FED-STD-H28
ASTM A564/A564M
ASTM B633
SAE AIR4127
SAE AMS5659
MS39210
SAE AMS5665
SAE AMS5862
SAE AMS2700
SAE-AMS-QQ-P-416
MS39212

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:48Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy-AS
Air Force-99
DLA - CC

Review activities:
Army-AR
Navy-MC,SA
Air Force-71

Preparing activity:
DLA - CC

(Project 4730-2012-038)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:48Z
Check the source to verify that this is the current version before use.